Advanced Functions & Equations

Section 1: Rational & Radical Inequalities

1 Solve the rational inequality: $\frac{x-3}{x+2} \leq 0$.

a. $(-2, 3]$

b. $[-2, 3]$

c. $(-\infty, -2) \cup [3, \infty)$

d. $x \leq 3$

2 What is the solution set of $\sqrt{2x+7} \geq x+2$?

a. $[-3.5, 1]$

b. $[-2, 1]$

c. $[-3.5, \infty)$

d. $[1, \infty)$

3 Solve $|2x-5| \leq 9$.

a. $[-2, 7]$

b. $[-7, 2]$

c. $[2, 7]$

d. $(-\infty, -2] \cup [7, \infty)$

Section 2: Partial Fraction Decomposition

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4 Decompose $\frac{5x-1}{x^2-1}$ into partial fractions $\frac{A}{x-1}+\frac{B}{x+1}$. What is the value of A?

a. 2

b. 3

c. 4

d. 5

5 In the partial fraction decomposition $\frac{7x+2}{(x-2)(x+4)}=\frac{A}{x-2}+\frac{B}{x+4}$, what is B?

a. 3

b. 4

c. 5

d. 6

6 What is the form of the partial fraction decomposition for $\frac{3x+1}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1}+\frac{Bx+C}{x^2+4}$

b. $\frac{A}{x-1}+\frac{B}{x^2+4}$

c. $\frac{A}{x-1}+\frac{B}{x+2}+\frac{C}{x-2}$

d. $\frac{Ax+B}{x-1}+\frac{C}{x^2+4}$

Section 3: Polynomial Roots & Graphing

7 According to Descartes' Rule of Signs, what are the possible numbers of positive real roots for $P(x)=x^4-3x^3+2x^2-5x+1$?

a. $4, 2,$ or 0

b. 3 or 1

c. 2 or 0

d. 1 only

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8 What is the multiplicity of the root $x = 3$ in the polynomial $P(x) = (x - 3)^3(x + 2)^2$?

a. 1

b. 2

c. 3

d. 5

9 What is the end behavior of the graph of $f(x) = -2x^5 + 4x^3 - x$?

a. As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$

b. As $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to -\infty$

c. Approaches $-\infty$ on both sides

d. Approaches ∞ on both sides

Section 4: Advanced Logarithmic & Exponential Modeling

10 Solve for x: $e^{2x} - 5e^x + 6 = 0$.

a. $x = \ln 2$ and $x = \ln 3$

b. $x = 2$ and $x = 3$

c. $x = e^2$ and $x = e^3$

d. No real solution

11 If $\log_3(x - 1) + \log_3(x + 7) = 2$, what is the value of x?

a. 2

b. 3

c. -8

d. 2 and -8

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12 Carbon-14 has a half-life of $5,730$ years. If a sample contains 25% of its original Carbon-14, how old is the sample?

a. $5,730$ years

b. $11,460$ years

c. $17,190$ years

d. $22,920$ years

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